MAHARSHI OZA

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PROFESSIONAL SUMMARY

Design Verification engineer with industry UVM experience and an MS in Computer Engineering (NC State). Skilled in SystemVerilog/UVM, constrained-random testing, SVA and functional coverage, with C/C++ test and tooling development, embedded ECU bring-up and debug, and low-power design (IEEE 1801/UPF, clock and power gating). Strong foundation in CPU microarchitecture, instruction sets and memory hierarchies.

CORE COMPETENCIES

Languages & Scripting: SystemVerilog, Verilog, C, C++, Python, Perl, Tcl, Shell, Makefile; AXI4, I2C, RISC-V ISA
Design Verification: UVM, Constrained-Random Verification, SVA, Functional & Code Coverage, Test Planning, Reusable Testbench Architecture, Scoreboards, TLM, Regression & Debug
Low Power & ASIC: IEEE 1801/UPF, Power Domains, Isolation, Level Shifters, Clock & Power Gating, RTL Design, Synthesis, STA, CDC, DFT
Computer Architecture: RISC-V, ARM Cortex-A/M (coursework), Out-of-Order Execution, Branch & Value Prediction, Caches, TLBs, Microbenchmarking
Tools: VCS, Xcelium, QuestaSim, Design Compiler, Conformal Low Power, Vivado, GDB, JTAG, Vector CANoe/CANalyzer, Git, Linux

PROFESSIONAL EXPERIENCE

RTL Design Verification Engineer | Maxgen Technologies Pvt Ltd., India **Aug 2023 – Aug 2024**

- Implemented an out-of-order scoreboard for an AXI4 memory subsystem in SystemVerilog/UVM, tracking outstanding transactions by ID and checking read/write data, responses and per-ID ordering.
- Developed functional coverage for AXI4 bursts, IDs, out-of-order completion and handshake stalls (over 90%); verified reset, FIFO full/empty and handshake corner cases and debugged regressions in Questa, Xcelium and VCS.
- Maintained UVM environments for Asynchronous FIFO and FPU blocks and contributed to multi-port packet router RTL (FSM control, FIFOs, synchronizers).

Functions Engineer – Vision & Perception | Magna Electronics, India **Dec 2021 – Aug 2023**

- Implemented requirements-based KPI checkers in C++ in the ADAS Lane Marker Detection application, turning SRD/SRS thresholds into self-checking pass/fail logic against ground truth, with code/branch/line coverage under MISRA C/C++.
- Validated classical and DNN (ONNX) lane-detection pipelines on recorded camera and CAN datasets, reusing the KPI checks to compare both.
- Supported ECU bench bring-up: deployed the C++ binary to target hardware, debugged it with GDB breakpoints and a JTAG debugger, and analyzed CAN signals in Vector CANoe/CANalyzer.

PROJECTS

UVM Verification of a Set-Associative Write-Back Cache Controller | SystemVerilog, UVM, SVA, Python

- Built a UVM 1.2 environment for a parameterized write-back/write-allocate cache controller with 3 agents (CPU, flush, reactive memory responder via TLM analysis FIFO), a reference-model scoreboard, and SVA protocol checks split into testbench-side and DUT-side assertions.
- Developed 7 directed and constrained-random tests using factory config overrides (backpressure, flush/CPU races, memory error injection); built a Python regression flow merging per-run coverage and reporting holes, with 0 data mismatches across 1,000+ accesses under injected errors.

Functional Verification of LC-3 Processor | UVM/UVMF

- Architected a reusable UVM/UVMF environment covering LC-3 pipeline stages and instruction/data memory; developed sequences, tests, assertions, regression, and coverage models for control-flow, datapath, and corner cases, achieving 100% functional coverage closure.
- Built independent scoreboard-based checking to validate instruction execution against expected pipeline behavior, enabling systematic bug isolation across fetch, decode, and execute stages.

UVM-Based Verification of AXI4 Slave | SystemVerilog

- Built a UVM environment for a 32-bit AXI4 slave supporting FIXED/INCR bursts; implemented agents, scoreboard, SVA protocol checks, and functional coverage for burst types, sizes, and boundary conditions, including constrained-random backpressure scenarios.

Low-Power Design with IEEE 1801 (Cadence course labs) | UPF, Conformal Low Power

- Wrote IEEE 1801 power intent (3 power domains, power switch, isolation cells, 0.9 V → 1.08 V level shifter) and checked it with Conformal Low Power.

Microarchitecture Reverse Engineering of x86 Processors | C, Python

- Wrote C microbenchmarks, built with Makefiles, to measure cache hierarchy, miss latencies, TLB and page-walk behavior, BTB capacity, ROB size and pipeline depth on Intel servers, with Python analysis of the results.

RISC-V Out-of-Order Simulator with Value Prediction | C++

- Built a cycle-accurate RISC-V out-of-order simulator with a stride value predictor and checkpoint recovery, improving harmonic-mean IPC from 1.91 to 2.00 across 15 benchmarks.

Pipelined Transformer Self-Attention Accelerator | SystemVerilog, Design Compiler

- Designed an SRAM-based self-attention accelerator (Q/K/V projections, score and attention matrices); verified against reference results with 217/217 checks passing across 4 test cases and closed timing at 5 ns in Nangate 45 nm.

FIR Filter Accelerator via High-Level Synthesis | SystemC, Catapult HLS, RISC-V Rocket-Chip

- Implemented a 16-tap FIR filter accelerator in SystemC and synthesized it with Catapult HLS at 5 ns, cutting latency from ~3.5K to 69 cycles.

EDUCATION

M.S. Computer Engineering | North Carolina State University, Raleigh, NC | CGPA: 3.7/4 **Aug 2024 – May 2026**
Relevant Coursework: Advanced Verification using UVM, ASIC Verification, Advanced Microprocessor Architecture, Microarchitecture Reverse Engineering
Training: Cadence – Fundamentals of IEEE 1801 Low-Power Specification Format (v12.0); Design for Test Fundamentals (v25.1)
B.Tech. Electronics & Communication Engineering | Dharmsinh Desai University, India **Aug 2018 – May 2022**